

Indian Statistical Institute

BSDS II

Stochastic Process

Class Test III (sample solution)

1. Consider the Markov Chain on $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$ with the transition matrix $\mathbf{P}$ given by

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{2}{5} & \frac{3}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{6} & \frac{1}{3} & 0 & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & 0 & \frac{1}{6} & 0 \end{bmatrix} \end{matrix}.$$

Find the values of f_{33}^* and f_{44}^* . What is the probability that the chain starting from 3, visits the state 4 before visiting state 6.

Sample Solution: We have (done in the class)

$$f_{33}^* = p_{33} + \sum_{j=1, j \neq 3}^6 p_{3j} f_{j3}^*.$$

We observe that $\{1, 2\}$ is a closed communicating class. So, $1 \not\leftrightarrow 3$ and $2 \not\leftrightarrow 3$. Thus, $f_{13}^* = f_{23}^* = 0$. Thus, the above equation becomes

$$f_{33}^* = p_{33} + \sum_{j=4}^6 p_{3j} f_{j3}^* = \frac{1}{6} + \frac{1}{3} f_{43}^* + \frac{1}{6} f_{63}^*$$

To compute f_{j3}^* for $j = 4, 5, 6$, we modify the transition matrix by making the state 3 as absorbing state. Thus, in the modified chain, we will have two closed classes, namely $\{1, 2\}$ and $\{3\}$. We note that f_{j3}^* for $j = 4, 5, 6$ is will be given by the probability of hitting closed class $\{3\}$. This is given by the minimal solution (here only one solution) of the equations

$$\begin{aligned} x_4 &= p_{43} + p_{44}x_4 + p_{45}x_5 + p_{46}x_6 = \frac{1}{8} + \frac{1}{8}x_4 + \frac{1}{8}x_5 + \frac{1}{8}x_6; \\ x_5 &= p_{53} + p_{54}x_4 + p_{55}x_5 + p_{56}x_6 = \frac{1}{6} + \frac{1}{6}x_4 + \frac{1}{6}x_5 + \frac{1}{6}x_6; \\ x_6 &= p_{63} + p_{64}x_4 + p_{65}x_5 + p_{66}x_6 = \frac{1}{8} + \frac{1}{8}x_5. \end{aligned}$$

Solving, we have $x_4 = \frac{9}{44}$, $x_5 = \frac{12}{44}$, and $x_6 = \frac{7}{44}$. Thus, $f_{43}^* = \frac{9}{44}$, $f_{53}^* = \frac{12}{44}$, $f_{63}^* = \frac{7}{44}$. Thus, $f_{33}^* = \frac{23}{88}$.

For the state 4, we have now

$$f_{44}^* = p_{44} + \sum_{j=3, j \neq 4}^6 p_{4j} f_{j4}^* = \frac{1}{8} + \frac{1}{8} f_{34}^* + \frac{1}{8} f_{54}^* + \frac{1}{8} f_{64}^*.$$

Following a similar method, i.e., making the state 4 absorbing and find the probability of hitting the closed class $\{4\}$ from 3, 5, 6. Therefore, we have the linear equations:

$$\begin{aligned} y_3 &= p_{34} + p_{33}y_3 + p_{35}y_5 + p_{36}y_6 = \frac{1}{3} + \frac{1}{6}y_3 + \frac{1}{6}y_6; \\ y_5 &= p_{54} + p_{53}y_3 + p_{55}y_5 + p_{56}y_6 = \frac{1}{6} + \frac{1}{6}y_3 + \frac{1}{6}y_5 + \frac{1}{6}y_6; \\ y_6 &= p_{64} + p_{63}y_3 + p_{65}y_5 + p_{66}y_6 = \frac{1}{8}y_3 + \frac{1}{8}y_5. \end{aligned}$$

Solving, $y_3 = \frac{79}{189}$, $y_5 = \frac{19}{63}$ and $y_6 = \frac{17}{189}$. Thus, $f_{34}^* = \frac{79}{189}$, $f_{54}^* = \frac{19}{63}$ and $f_{64}^* = \frac{17}{189}$. Hence, we have $f_{44}^* = \frac{1}{8} + \frac{1}{8}f_{34}^* + \frac{1}{8}f_{54}^* + \frac{1}{8}f_{64}^* = \frac{19}{84}$.

To find the probability of hitting the state 4 before 6, we make both the states 4 and 6 as absorbing. Therefore, in the modified chain, we will have 3 closed classes, namely $\{1, 2\}$, $\{4\}$ and $\{6\}$. We now want to find the probability of hitting the closed class $\{4\}$ from the states 3 and 5. Again, this is given by the minimal (here unique) solution of the system of linear equation

$$\begin{aligned} z_3 &= p_{34} + p_{33}z_3 + p_{35}z_5 = \frac{1}{3} + \frac{1}{6}z_3; \\ z_5 &= p_{54} + p_{53}z_3 + p_{55}z_5 = \frac{1}{6} + \frac{1}{6}z_3 + \frac{1}{6}z_5. \end{aligned}$$

Solving, we have $z_3 = \frac{2}{5}$ and $z_5 = \frac{7}{25}$. Thus, the required probability is $\frac{2}{5}$.

2. Consider the Markov Chain on $\mathcal{S} = \{1, 2, 3, 4\}$ with the transition matrix $\mathbf{P}$ given by

$$\mathbf{P} = \begin{array}{c} \begin{array}{cccc} & 1 & 2 & 3 & 4 \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{2}{5} & \frac{3}{5} & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array} \end{array}.$$

Find all the stationary distributions.

Sample Solution: Here we have 3 communicating classes $\mathcal{C}_1 = \{1, 2\}$, $\mathcal{C}_2 = \{4\}$ and $\mathcal{C}_3 = \{3\}$. Note that the classes $\mathcal{C}_1$ and $\mathcal{C}_2$ are closed. Since the state space is finite, all closed classes are positive recurrent. Therefore, when restricted to closed class, there will be a unique stationary distribution and all stationary distributions are convex combinations of the these stationary distributions. So, for the class $\mathcal{C}_1$, let $\boldsymbol{\pi}_1 = (\pi(1), \pi(2))$ be a stationary probability distribution, i.e.,

$$\pi(1) + \pi(2) = 1 \text{ and } \boldsymbol{\pi} \tilde{\mathbf{P}} = \boldsymbol{\pi}$$

where $\tilde{\mathbf{P}}$ is the transition matrix restricted to the states $\{1, 2\}$, i.e., $\tilde{\mathbf{P}} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{2}{5} & \frac{3}{5} \end{bmatrix}$. Writing the equations, we have

$$\begin{aligned} \pi(1) &= \pi(1)p_{11} + \pi(2)p_{21} = \frac{1}{2}\pi(1) + \frac{2}{5}\pi(2); \\ \pi(2) &= \pi(1)p_{12} + \pi(2)p_{22} = \frac{1}{2}\pi(1) + \frac{3}{5}\pi(2) \end{aligned}$$

and $\pi(1) + \pi(2) = 1$. Note even though there are 3 equations with two unknowns, the first two equations are actually the same as they both reduce to $\pi(1) = \frac{4}{5}\pi(2)$. Solving we get $\pi(1) = \frac{4}{9}$ and $\pi(2) = \frac{5}{9}$.

For the closed class $\mathcal{C}_2 = \{4\}$, the transition matrix restricted to the state $\{4\}$ is $[1]$. Therefore, the only probability $\boldsymbol{\pi}_2 = (\pi(4))$ is $\boldsymbol{\pi}_2 = (1)$.

We now extend each of these probabilities to the whole state space by giving 0 probability to other states, i.e., $\widetilde{\boldsymbol{\pi}}_1 = (\frac{4}{9}, \frac{5}{9}, 0, 0)$ and $\widetilde{\boldsymbol{\pi}}_2 = (0, 0, 0, 1)$. Therefore, all stationary distributions for this chain given by $\lambda \widetilde{\boldsymbol{\pi}}_1 + (1 - \lambda) \widetilde{\boldsymbol{\pi}}_2 = (\frac{4\lambda}{9}, \frac{5\lambda}{9}, 0, (1 - \lambda))$ for any $\lambda \in [0, 1]$.